Gongati Adinesh Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.30
Sri Chaitanya Junior College,Hyderabad	Class XII (Board of Intermediate Education,TG)	2024	97.60%
DR. K K R's Gowtham EM School,Gudivada	Class X (Board of Secondary Education,AP)	2022	88.83%

POSITIONS OF RESPONSIBILITY

Multimedia Coordinator at **PRAYAS RDC CLUB, IIT HYDERABAD**

- Managed multimedia operations, including videography, video editing, and social media content creation, while volunteering across multiple PRAYAS events to improve outreach, event documentation, and online engagement

SKILLS

Programming Languages: C, C++, JavaScript, SQL ,RISC-V(assembly).

Web Technologies: HTML, CSS, ReactJS.

Databases: MySQL, PostgreSQL.

Operating Systems: Ubuntu, Windows.

Others: Git, Github, Vim, Solid Edge.

RELEVANT COURSES

Computer Science: Introduction to Programming, Discrete Mathematics, Software Development Fundamentals, Operating Systems, Computer Architecture, Database Management Systems, Data Structures, Algorithms, Theory of Computation, Introduction to Compilers.

Mathematics: Calculus, Linear Algebra, Probability and Random Variables, Transform Techniques, Introduction to Statistics.

Others: Introduction to Artificial Intelligence, Digital Circuits, Digital Fabrication, Communication Skills.

PROJECTS

Experience with xv6 Operating System | C, xv6 (RISC-V)

- Extended the xv6 kernel by implementing **custom system calls** and per-process system call accounting, improving kernel functionality and enabling detailed monitoring of system call usage.
- Replaced the default **Round Robin** scheduler with a **Multi-Level Feedback Queue (MLFQ)** scheduler supporting **priority-based preemption**, **configurable time slices**, and **global priority boosting** for starvation prevention.
- Built a **virtual memory subsystem** using **page-fault-driven lazy allocation**, the **Clock page replacement** policy, and a **disk-backed swap space** with **swap-in** and **swap-out** support, enabling processes to execute beyond memory limits.
- Developed configurable **disk scheduling** for **swap I/O**, integrated **RAID (0/1/5)**-based multi-disk swap storage for

disk failure recovery, and implemented **kernel statistics** to analyze disk and swap activity across different workloads.

Rubik's Cube Solver | C++ (oops)

- Designed a modular Rubik's Cube solver in C++ using **Object-Oriented Programming**, supporting interchangeable **1D and 3D cube representations** through a common abstract interface for model-independent operations.
- Implemented four **model-independent search algorithms**--**DFS, BFS, IDDFS, and IDA***--with efficient hash-based state storage to solve scrambled cube configurations while eliminating duplicate state exploration.
- **Optimized** the **IDA*** solver using a **Corner Pattern Database** and **Nibble Array** for compact heuristic storage, reducing search time and memory usage while outperforming the **DFS, BFS, and IDDFS** implementations.

Arbitrary Precision Arithmetic Library | Java

- Built a Java library for **arbitrary-precision integer and floating-point arithmetic** by representing numbers as **strings**, eliminating the size and precision limitations of native data types.
- **Implemented addition, subtraction, multiplication, and division** with support for **30-digit decimal precision**, preserving numerical accuracy while preventing overflow and rounding errors through manual arithmetic operations.
- Packaged the implementation into a reusable Java package and executable JAR, allowing developers to integrate arbitrary-precision arithmetic into their applications.

Computer Architecture Optimization Using FPU Registers | RISC-V Assembly

- Optimized RISC-V matrix multiplication by using FPU registers as temporary storage instead of repeated load/store instructions, reducing memory access overhead during computation.
- **Implemented** a **5×5 blocked matrix multiplication** algorithm in **RISC-V Assembly** and evaluated its performance against a baseline implementation using **Gem5** and **RARS** simulators.
- **Reduced** memory accesses by **75%** and analyzed the performance improvements through **Gem5**, demonstrating the impact of register-level optimization on execution efficiency.

EXTRACURRICULAR

- Member of the **National Service Scheme (NSS).**

- **Hobbies & Interests |** Cricket, Movies, Music.